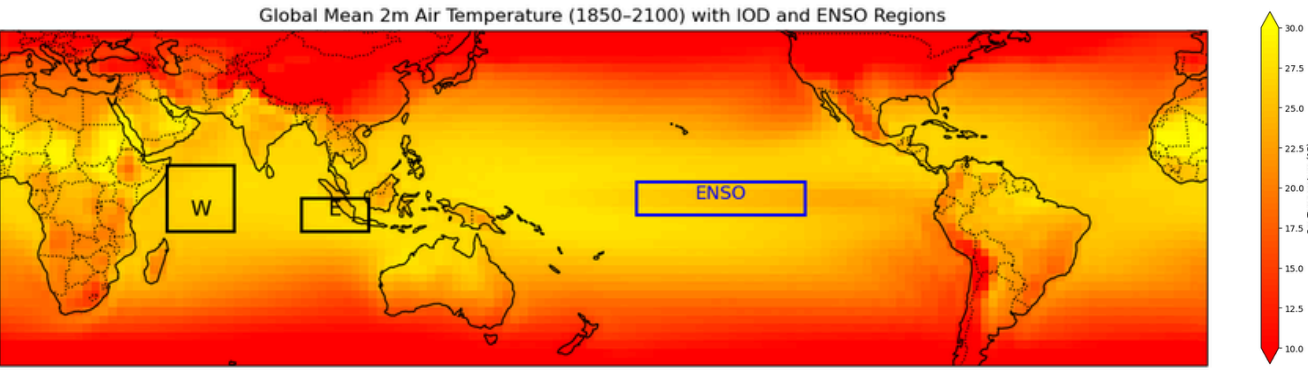


Global Temperature Teleconnections of ENSO and IOD: A Comparative Phase-Based Analysis

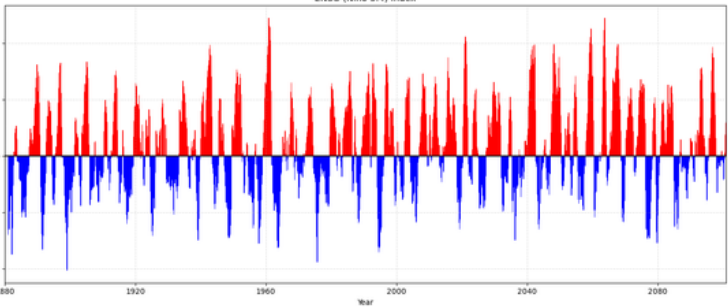
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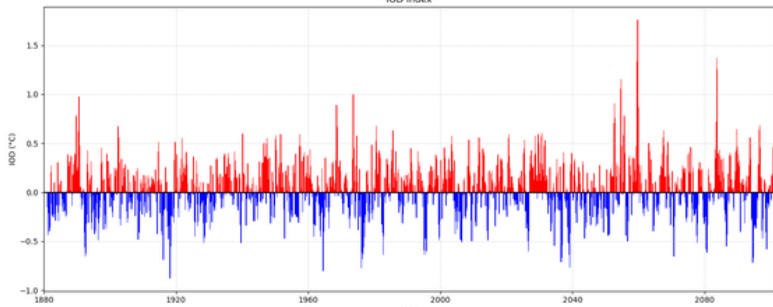
The **Indian Ocean Dipole (IOD)** and the **El Niño–Southern Oscillation (ENSO)** are two dominant coupled ocean-atmosphere phenomena occurring in the Indian and Pacific Oceans, respectively. These climate drivers play a crucial role in modulating weather and climate patterns on a global scale. Both events are known for their significant influence on temperature, rainfall, and extreme events in various regions including East Africa, India, Australia, and the United States via atmospheric teleconnections that extend far beyond their oceanic origins (Saji et al., 1999). In this study, we investigate how IOD and ENSO phases are regionally linked to global surface air temperature anomalies. By focusing on spatial correlations and composite mean analyses, we aim to better understand their individual and overlapping impacts on remote regions, shedding light on the geographic reach of these oceanic patterns.



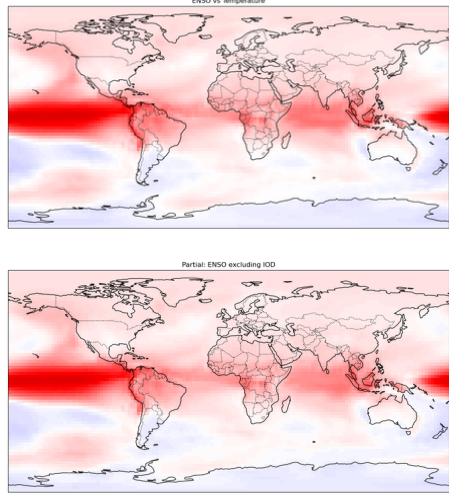
From Past to Future: A Timeline of ENSO and IOD



- ENSO and IOD indices are derived from ICON model output (tas_Amon_MPI-ESM1-2-LR_historical-ssp245) using 2 m air temperature as a proxy for SST.
- IOD is calculated as the anomaly difference between two Indian Ocean regions (see Fig. 1).
- ENSO (Niño 3.4) is computed as the average anomaly over the central Pacific region (see Fig. 1).
- A 30-year moving climatology and 3-month rolling mean are applied to smooth the index time series.
- Figures 2 and 3 show the final ENSO and IOD indices highlighting temporal variability.

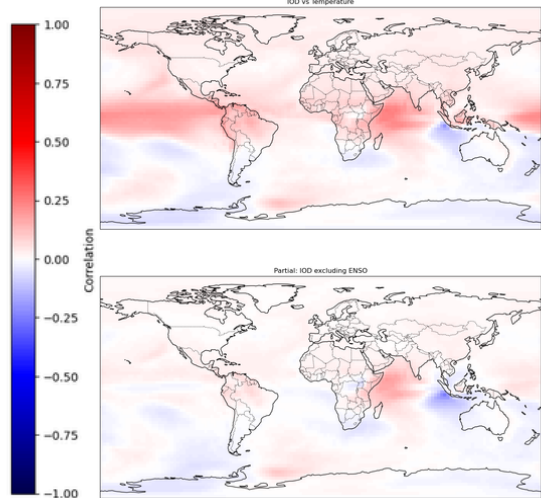


Separating ENSO and IOD Footprints on Global Temperature



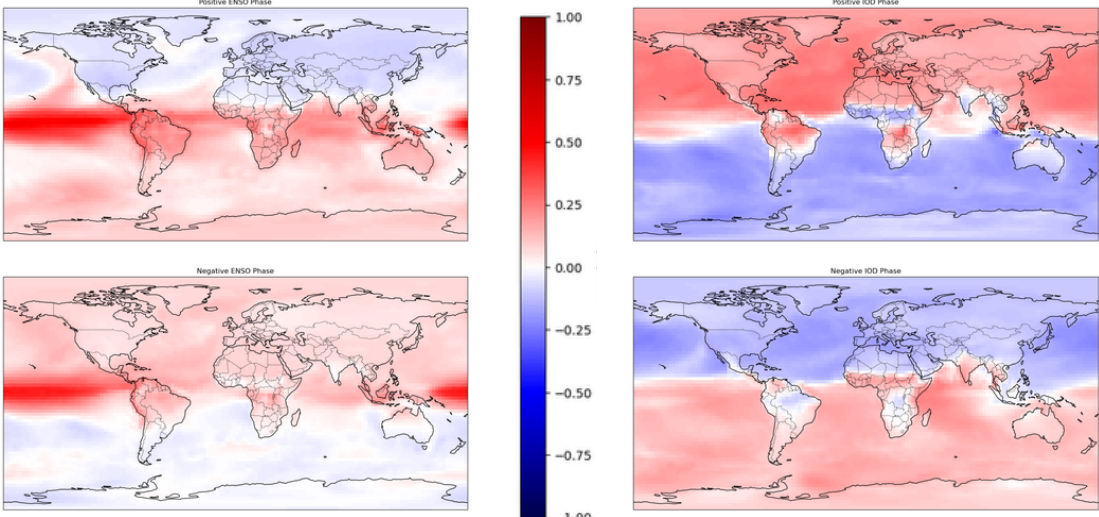
- Figure 6 illustrates the global correlation pattern between ENSO and surface temperature, highlighting strong positive correlations across the equatorial Pacific and extending into the Americas, southern Africa, and parts of Asia, confirming ENSO's major global teleconnections.
- Figure 7 presents the partial correlation of ENSO with temperature while removing IOD influence. The spatial pattern remains largely intact, indicating that ENSO's influence is robust and not significantly confounded by the IOD.
- Overall, ENSO exhibits strong and widespread climatic impacts, particularly across the Pacific Basin and adjacent land regions, affirming its dominant role in modulating global temperature variability.

- Limited Global Reach: The full correlation map (Fig. 4) shows that while IOD strongly affects temperatures around the Indian Ocean, its global teleconnection pattern is relatively weak, especially compared to ENSO.
- ENSO Overlap Removed: After controlling for ENSO (Fig. 5), the remaining IOD signal becomes more localized, primarily around eastern Africa and the southeastern Indian Ocean, confirming ENSO's dominant confounding influence.
- Regional Specificity: The residual IOD signal appears to maintain influence in a few regions, notably the East African Rift and western Maritime Continent, suggesting regional relevance even if global significance is limited.



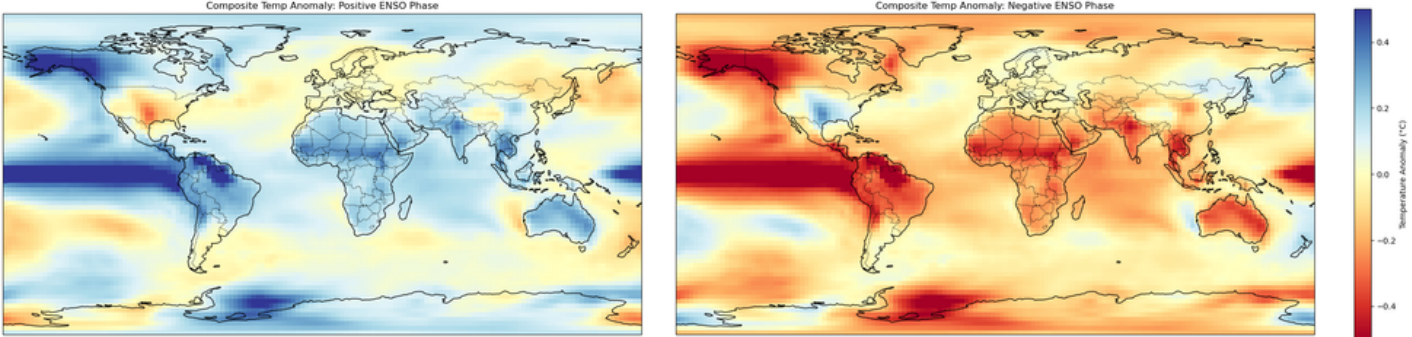
Unpacking the Bipolar Nature of ENSO and IOD

- Opposite Hemispheric Responses: The positive ENSO phase (El Niño) is associated with warming in the southern tropics and cooling in the northern mid-latitudes, while the negative phase (La Niña) shows the opposite pattern highlighting a strong hemispheric asymmetry in ENSO's global influence.
- Stronger Pacific Influence in El Niño: Positive ENSO (El Niño) shows a dominant warming signal across the central and eastern tropical Pacific, confirming well-established teleconnections with South America, Africa, and Australia, consistent with findings from [Ropelewski & Halpert, 1987].
- Amplified Land-Tropics Coupling in La Niña: Negative ENSO phases enhance warming across the Indo-Pacific and tropical land regions such as Central Africa and the Maritime Continent, likely due to intensified Walker circulation and latent heat feedbacks ([Trenberth et al., 1998]).



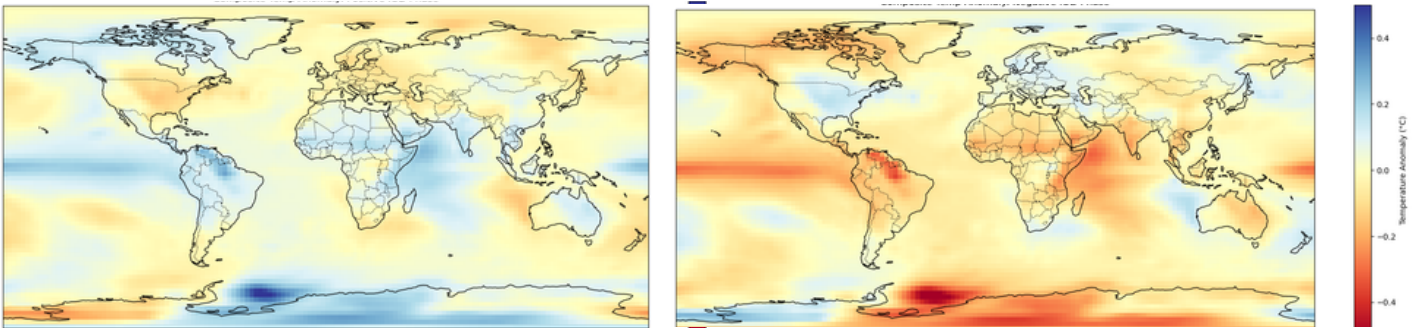
- Clear Hemispheric Flip: The positive IOD phase shows strong warming correlations over the Northern Hemisphere, while the negative phase shifts this influence to the Southern Hemisphere. This hemispheric flip suggests distinct atmospheric teleconnection patterns depending on the IOD sign.
- Thermal Gradient and Walker Circulation Shifts: This flip may arise from shifts in the Walker and Hadley circulations due to IOD-induced changes in equatorial heat gradients. Positive IOD cools the eastern Indian Ocean and warms the west, intensifying the Walker circulation and extending its influence northward (Cai et al., 2014, Nature). The opposite holds for negative phases.
- Implications for Seasonal Forecasting: These opposing patterns imply that accurate phase prediction of IOD is crucial for seasonal temperature forecasting, especially across continents like Asia, Africa, and South America, where impacts are seasonally sensitive and phase-dependent.

Climatic Imprints of ENSO and IOD Events



- Clear Global Signal: The positive ENSO phase (El Niño) is associated with widespread cooling across tropical landmasses and warming in the eastern and central Pacific. Conversely, the negative ENSO phase (La Niña) shows warming over much of the tropics, especially South America and Africa, indicating ENSO's dominant control over global temperature variability.
- Zonal Response Pattern: A pronounced east-west dipole structure appears over the Pacific and surrounding regions, reflecting the zonally extensive influence of ocean-atmosphere coupling during ENSO events. This pattern extends into the Atlantic and Indian Oceans, illustrating ENSO's ability to project teleconnections globally.
- Amplified Continental Effects: Regions such as the Amazon basin, equatorial Africa, and Southeast Asia show strong temperature contrasts between phases. These effects align with established ENSO teleconnection pathways that influence regional circulation and cloud-radiation feedbacks (Trenberth et al., 2002).

- Weaker Global Signal Compared to ENSO: Both positive and negative IOD phases show weaker global composite temperature anomalies than ENSO, indicating a more localized influence primarily around the Indian Ocean and adjacent landmasses.
- Polar Amplification Response: Positive IOD phases (right panel) tend to correspond with modest warming across the Northern Hemisphere mid-to-high latitudes and cooling near the Southern Ocean. The reverse pattern is seen for negative IOD phases, suggesting possible links to hemispheric asymmetries in atmospheric circulation and polar amplification processes (e.g., Abram et al., 2020).
- Land-Ocean Contrast in Teleconnections: Temperature anomalies over continents like Africa and South America are evident but relatively subtle. This further supports the limited remote teleconnections of IOD compared to ENSO, which is consistent with earlier studies showing IOD-driven temperature effects are spatially constrained and often modulated by concurrent ENSO phases (e.g., Cai et al., 2011).



Conclusions

- ENSO exhibits a significantly broader and stronger regional influence on global temperature variability compared to the IOD. This is evident from higher spatial correlations and more coherent teleconnection patterns across the tropics and mid-latitudes.
- The positive and negative phases of both ENSO and IOD display hemispheric asymmetry in their temperature impacts. This is largely due to seasonally modulated shifts in the Intertropical Convergence Zone (ITCZ) and changes in Hadley circulation, which redistribute heat differently across hemispheres during opposite phases (Chiang & Friedman, 2012).
- Regions with the strongest correlations—such as the tropical Pacific for ENSO and the western Indian Ocean for IOD—highlight areas of direct climatic influence. These hotspots indicate where ocean-atmosphere coupling exerts the most substantial control on surface air temperature anomalies.
- Partial correlation and composite analyses reveal that while ENSO maintains a dominant global signal independently, IOD's influence is more localized and often intertwined with ENSO variability. This suggests IOD acts more as a regional modulator than a standalone global driver.

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